Notes: Cuñãt (RFS, 2007)

Trade Credit: Suppliers as Debt Collectors and Insurance Providers

Contents

1 One-sentence summary

Cunãt explains costly trade credit as a natural outcome of supplier-customer relationships: suppliers have a collection advantage because they can threaten to stop supplying intermediate goods, and they also provide liquidity insurance through late payment and maturity extensions.

2 Big picture

2.1 The trade credit puzzle

Trade credit is expensive when measured by early-payment discounts, yet it is widely used. A standard "2-10 net 30" contract implies a very high annualized interest rate, and trade credit is a large share of short-term finance for many firms.

- **Puzzle 1:** Why do firms borrow from suppliers at apparently high rates instead of using bank credit?
- **Puzzle 2:** Why do suppliers lend to troubled firms when banks are reluctant to lend?
- **Puzzle 3:** Why is late payment so common, and why is it often tolerated without explicit penalties?

2.2 Central answer

Suppliers have a comparative advantage in lending because they are embedded in the production relationship. They can punish default by stopping the supply of intermediate inputs, and they value preserving the relationship when temporary liquidity shocks threaten the customer.

2.3 Two roles of suppliers

- **Debt collector:** The supplier's threat to stop supplying specific inputs makes repayment more enforceable than it is for a bank.
- **Insurance provider:** The supplier may help customers survive liquidity shocks because preserving the relationship creates future surplus.

2.4 What looks like an expensive loan

The high implicit interest rate on trade credit includes three components:

1. the standard funding cost;
2. a default premium;
3. an insurance premium for future liquidity support.

2.5 Main empirical prediction

Trade credit should be higher for firms with weak collateral, low liquidity, temporary liquidity shocks, high growth, poor performance, and young supplier-customer relationships where the relationship has begun to develop but is not yet fully mature.

3 Incremental contribution of the paper

3.1 A relationship-based theory of trade credit

Incremental contribution: The paper shifts the explanation of trade credit from transaction convenience or pure price discrimination to the enforceability and insurance properties of supplier-customer relationships.

3.2 Supplier collection advantage over banks

The model gives suppliers a credible repayment threat that banks lack: cutting off future inputs. This makes supplier debt more enforceable precisely when the customer uses relationship-specific intermediate goods.

3.3 Trade credit as liquidity insurance

The paper interprets late payment and extension of trade credit as an implicit insurance arrangement. Suppliers help customers in temporary distress because liquidating the relationship destroys future surplus.

3.4 A pricing explanation for high implicit rates

The paper explains why measured trade credit rates can be high without requiring irrational borrowers or monopoly pricing. The interest rate must compensate suppliers for default risk and for providing liquidity support.

3.5 Empirical link to firm conditions

The paper connects the theory to firm-level evidence from UK firms. It tests whether trade credit rises when firms are financially constrained, growing quickly, performing poorly, or in early stages of their life cycle.

3.6 Why this contribution matters

The paper shows that trade credit is not a second-best substitute for bank credit. It is a different financial contract generated by the real input relationship between supplier and buyer.

4 Conceptual framework

4.1 Supplier-customer surplus

A key primitive is relationship surplus. A supplier and customer produce more value together once the supplier becomes specific to the customer or once the production relationship matures.

4.2 Limited enforceability

Bank debt is limited by collateral and by legal enforceability. Suppliers can lend beyond the collateral limit because they can threaten to stop supplying the customer.

4.3 Liquidity shocks

Customers can experience temporary liquidity shocks that are socially worth covering. If the customer survives, the relationship continues and the supplier earns future surplus.

4.4 Late payment as an option

Late payment is interpreted as an implicit option granted by the supplier. Customers may pay after the due date when liquidity shocks arrive, and the supplier tolerates this if preserving the relationship is valuable.

4.5 Cost of funds amplification

If suppliers face a higher cost of funds than banks, they pass this cost through to customers. Because trade credit includes default and insurance premia, a higher supplier funding cost is amplified in the implicit trade credit rate.

5 Model setup and derivations

5.1 Agents and production

There are banks, suppliers, and customers. Customers need intermediate inputs from suppliers. A customer can be in a startup stage or a mature stage. The mature relationship creates surplus and gives suppliers enforcement power.

5.2 Bank credit and collateral

Banks lend only against collateral. If I_t is project scale and κ is pledgeable collateral, the bank lending constraint can be written schematically as

$$B_t \leq \kappa I_t.$$

Detailed interpretation: This constraint captures why bank lending is insufficient for firms with little collateral or temporary liquidity needs.

5.3 Supplier enforcement

The supplier can enforce repayment because default destroys the relationship and returns the customer to a less productive search or startup stage. The maximum enforceable supplier repayment is therefore tied to the continuation value of the relationship.

5.4 Trade credit interest rate

The model writes the implicit trade credit interest rate as a funding-cost term multiplied by default and insurance premia. In schematic form,

$$1 + r^{TC} = (1 + r)\left[1 + \text{default premium} + \text{insurance premium}\right].$$

If suppliers face funding cost $r^s > r$, then

$$1 + r^{TC} = (1 + r^s)\left[1 + \text{default premium} + \text{insurance premium}\right].$$

Detailed interpretation: This is the paper's key pricing result: trade credit appears expensive because it embeds default risk and liquidity insurance.

5.5 Liquidity support

When a liquidity shock arrives, the supplier may pay the shock cost to keep the customer alive. The supplier does so when

$$\text{continuation value of relationship} > \text{cost of liquidity support.}$$

Detailed interpretation: This gives a microfoundation for late payment. The supplier does not simply fail to collect; the supplier optimally preserves a valuable relationship.

5.6 Bank insurance versus supplier insurance

A third party such as a bank could in principle insure the liquidity shock. But supplier insurance dominates when customers cannot commit to buy bank insurance and when suppliers are already tied to the continuation value of the relationship.

6 Data and measurement

6.1 Main dataset

The empirical work uses FAME, a Bureau van Dijk database covering UK firms over 1993-2002. The paper focuses on firm-level panel variation in balance-sheet measures.

6.2 Trade credit measures

The two main dependent variables are:

$$\frac{\text{Trade credit}}{\text{Assets}}, \qquad \frac{\text{Trade credit}}{\text{Total debt}}.$$

The first captures the scale of trade credit relative to firm assets. The second captures the importance of trade credit relative to other debt sources.

6.3 Financial constraint proxies

The key explanatory variables include:

- collateral, measured as fixed assets over total assets;
- liquid assets, measured as cash and liquid deposits over total assets;
- inventories;
- cost of sales;
- size;
- liquidity-shock indicators.

6.4 Liquidity shock measures

Two shock variables are used:

- **Liquid assets drop**: equals one if liquid assets/assets fall by more than 10 percentage points;
- **Dividend cut**: equals one if dividends fall from the previous year.

6.5 Performance and relationship measures

The paper uses asset growth rates as a performance measure and firm age as a proxy for the length of the supplier-customer relationship.

7 Empirical specification and identification design

7.1 Baseline panel regressions

The basic fixed-effect specification is

$$Y_{it} = \alpha_i + \lambda_t + \beta_1 Collateral_{it} + \beta_2 LiquidAssets_{it} + \Gamma X_{it} + \varepsilon_{it},$$

where Y_{it} is a trade-credit measure, α_i are firm fixed effects, and λ_t are year effects.

7.2 Liquidity shock regressions

The liquidity-shock specification replaces liquid assets with shock indicators:

$$Y_{it} = \alpha_i + \lambda_t + \delta_1 LiquidAssetsDrop_{it} + \delta_2 DividendCut_{it} + \Gamma X_{it} + \varepsilon_{it}.$$

7.3 Spline regressions

To test nonlinear predictions, the paper uses splines in asset growth and firm age. These regressions allow the slope of trade credit use to vary across ranges of growth or age.

7.4 Nonparametric regressions

The figures show local linear regressions of trade credit on growth and age. These are useful because the theory predicts nonlinear patterns: U-shapes and humps are more informative than simple average slopes.

7.5 Identification-style interpretation

The paper is not based on a randomized shock. The empirical design uses within-firm variation over time, firm fixed effects, year dummies, controls, and nonlinear patterns to test predictions of the relationship-based trade credit model.

7.6 Main threats

Potential concerns include omitted time-varying firm conditions, measurement error in trade credit, simultaneous changes in bank credit availability, and imperfect proxies for supplier-customer relationship strength. The paper addresses these with fixed effects, alternative dependent variables, shock proxies, GMM specifications, and nonlinear tests.

8 Tables and figures

8.1 Table 1: Relative size of trade credit; Table 2: Summary statistics

Read:

- Table 1 shows that trade credit is economically large: 25% of assets and 47% of short-term debt in the UK FAME sample; 17% of assets and 50% of short-term debt in the US NSSBF sample.
- Table 2 shows that in the UK firm-year panel, average trade credit/assets is 0.25 and average trade credit/total debt is 0.41.
- Collateral averages 0.25 and liquid assets average 0.12, giving cross-sectional variation in bank-credit capacity and liquidity.

Economic message: Trade credit is not a minor accounting item. It is a major source of short-term finance, so a theory of trade credit is essential for understanding firm financing.

Detailed interpretation: Table 1 motivates the puzzle; Table 2 defines the empirical environment. The paper then asks why firms hold so much trade credit despite apparently high implicit interest rates.

Table 1
Relative size of trade credit (average over firms)

Country	Data set	Trade credit/assets (%)	Trade credit/total debt (%)	Trade credit/Short term debt (%)
United Kingdom	FAME	25	41	47
United States	NSSBF	17	35	50

FAME, Bureau Van Dijk database contains data on over 250,000 UK firms of all sizes; NSSBF, National Survey on Small Business Finance 4630 small and medium US firms.

(a) Table 1: relative size of trade credit.

Table 2
Summary statistics

Variable	Mean	SD	q 10%	Median	q 90%
Trade credit/assets	0.25	3.54	0.03	0.17	0.49
Trade credit/total debt	0.41	0.25	0.09	0.38	0.78
Size	7.30	2.18	3.68	7.51	10.78
Inventories	0.23	0.18	0.03	0.19	0.48
Collateral	0.25	0.20	0.02	0.20	0.55
Liquid assets	0.12	0.17	0	0.05	0.35
Cost of sales	1.89	24.84	0.48	1.28	3.14
Growth	0.21	7.43	−0.18	0.05	0.45
Dividend cut	0.07	0.26	0	0	0
Liquid assets drop	0.06	0.23	0	0	0

(b) Table 2: summary statistics.

8.2 Table 3: Basic regressions; Table 4: Liquidity shocks

Read:

- Table 3 shows that collateral is negatively related to both trade credit/assets and trade credit/total debt.
- Firms with fewer pledgeable assets rely more on supplier finance.
- Liquid assets are negatively related to trade credit/assets and positively related to trade credit/total debt, consistent with liquidity and debt-composition effects.
- Table 4 shows that liquidity shocks, especially liquid asset drops, increase the use of trade credit.

Economic message: Suppliers fill financing gaps when bank-credit capacity and liquidity are weak. This supports the supplier-as-liquidity-provider channel.

Detailed interpretation: The negative coefficient on collateral is the core enforcement result. Firms with fewer assets to pledge to banks have more trade credit. Liquidity shocks show that trade credit also responds to temporary financing needs.

Table 3
Basic regressions

	Trade credit/assets			Trade credit/total debt		
Dependent variables	1	2	3	4	5	6
Lagged dependent variable			0.123 (1.88)*			0.352 (24.89)**
Collateral	−0.1343 (43.91)***	−0.1336 (43.68)***	−0.148 (24.17)***	−0.3101 (44.22)***	−0.311 (44.35)***	−0.0377 (24.80)
Liquid assets	−0.0948 (34.68)***	−0.0947 (34.65)***	−0.074 (14.62)***	0.1134 (14.32)***	0.1121 (14.17)***	0.048 (2.95)**
Inventories	0.0749 (22.04)***	0.0748 (22.02)***	0.086 (10.66)***	0.0923 (10.42)***	0.0926 (10.46)***	0.08 (4.41)***
Cost of sales	0.0079 (29.17)***	0.0078 (28.92)***	0.004 (2.34)**	0.0087 (14.99)***	0.0087 (14.99)***	0.0051 (1.29)
Size	−0.0137 (20.11)***	−0.0138 (20.21)***	−0.001 (0.50)	−0.0277 (16.77)***	−0.0284 (17.17)***	−0.047 (8.65)**
Sector-specific trends	No	Yes	No	No	Yes	No
Observations	170,600	170,600	110,016	92,339	92,339	49,391
Number of firms	37,130	37,130	26,468	24,155	24,155	14,375
R^2	0.07	0.07	—	0.05	0.06	—

Collateral, fixed assets/total assets; cost of sales, total cost of sales/total assets; inventories, finished and work in progress stocks/total assets; liquid assets, cash and liqu deposits/total assets; size, log of total assets.
Absolute value of t-statistics in parentheses. All regressions include as control variables: firm-fixed effects, year dummies, asset growth rates and age (not reported). Column and 4 include sector-specific time trends, and sectors are defined at three-digit Standard Industrial Clasification (SIC) level. Columns 3 and 6 use the GMM Arellano–Bo estimation procedure. GMM estimator uses first differences and includes up to three lags as instrumental variables t-statistic in parenthesis (z-statistic for GMM).
*, **, and *** significant at 10, 5, and 1%, respectively.

(a) Table 3: basic fixed-effect and GMM regressions.

Table 4
Liquidity shocks

	Trade credit/assets		Trade credit/total debt	
Dependent variables	1	3	2	4
Liquid assets drop	—	0.0035 (4.02)***	—	0.0109 (4.87)**
Dividend cut	−0.0006 (0.82)	—	0.0036 (2.20)**	—
Collateral	−0.1054 (40.72)***	−0.1060 (40.90)***	−0.3260 (55.82)***	−0.3271 (55.97)*
Inventories	0.1122 (38.76)***	0.1115 (38.46)***	0.0484 (6.51)***	0.0472 (6.35)**
Cost of sales	0.0013 (14.01)***	0.0013 (14.02)***	0.0104 (18.26)***	0.0104 (18.26)*
Size	−0.0182 (31.17)***	−0.0182 (31.19)***	−0.0231 (16.43)***	−0.0231 (16.43)*
Observations	205,780	205,780	113,099	113,099
Number of firms	39,599	39,599	26,468	26,468
R^2	0.05	0.05	0.05	0.05

Collateral, fixed assets/total assets; cost of sales, total cost of sales/total assets; dividend cut, dummy if dividends fell from last year nominal level; inventories, finished and wo in progress stocks/total assets; liquid assets drop, dummy if liquid assets/total assets fell by more than 10% last period; liquid assets, cash and liquid deposits/total assets; size, l of total assets.
Absolute value of t-statistics in parentheses. All regressions include as control variables: firm-fixed effects, year dummies, asset growth rates, and age (not reported).

(b) Table 4: liquidity shocks and trade credit.

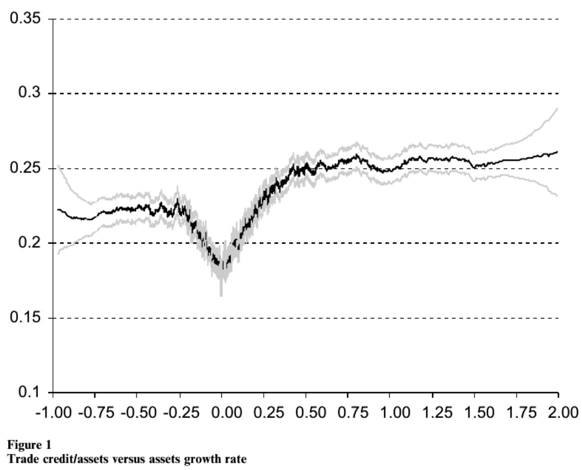
8.3 Figure 1: Trade credit/assets versus assets growth rate; Figure 2: Trade credit/total debt versus assets growth rate

Read:

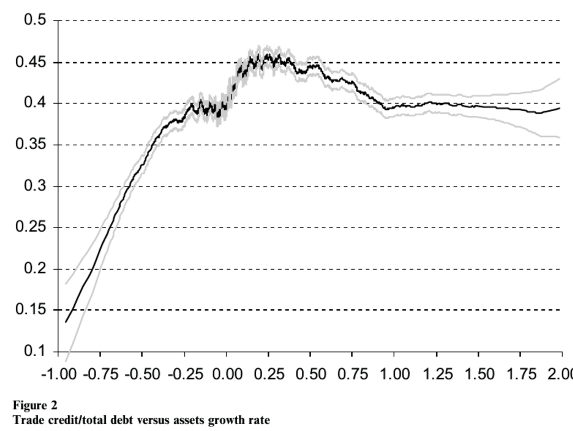
- Figure 1 shows a U-shaped relation between trade credit/assets and asset growth.
- Trade credit/assets is high for rapidly growing firms and also elevated for poorly performing firms.
- Figure 2 shows that trade credit/total debt rises sharply as firms move from very negative growth toward positive growth, then stabilizes at high levels.

Economic message: Firms use trade credit both when they grow quickly and need funds, and when they perform poorly and need liquidity support. This is exactly the dual role of suppliers as financiers and insurers.

Detailed interpretation: The U-shape is important because a linear regression would miss the two-sided nature of trade credit demand. Supplier finance is valuable in both distress and rapid expansion.



(a) Figure 1: trade credit/assets versus asset growth.



(b) Figure 2: trade credit/total debt versus asset growth.

8.4 Table 5: Performance splines; Table 6: Age splines

Read:

- Table 5 converts the nonlinear growth patterns into spline regressions with controls.
- The growth splines confirm higher trade credit at high growth rates and support the nonmonotonic pattern.
- Table 6 shows that trade credit/assets rises strongly during the first year of firm life and then declines, while trade credit/total debt rises in early years and stabilizes.

Economic message: Table 5 supports the liquidity/growth channel; Table 6 supports the relationship channel. Trade credit rises after suppliers and customers begin to build a relationship, but older firms rely less on it relative to assets.

Detailed interpretation: The age result helps distinguish pure financial distress from relationship enforcement. Very young firms do not yet have fully developed supplier relationships, but as relationships develop trade credit rises.

Table 5
Performance splines

	Trade credit/assets	Trade credit/total debt
Growth spline (–1 to –0.75)	0.1124 (1.27)	0.0844 (0.29)
Growth spline (–0.75 to –0.5)	0.0038 (0.15)	0.2984 (4.05)***
Growth spline (–0.5 to –0.25)	–0.0152 (1.51)	0.0889 (3.44)***
Growth spline (–0.25 to 0)	0.0296 (6.54)***	0.0869 (8.45)***
Growth spline (0–0.25)	0.0598 (16.43)***	0.0280 (3.58)***
Growth spline (0.25–0.5)	0.0281 (4.87)***	–0.0050 (0.40)
Growth spline (0.5–0.75)	–0.0042 (0.51)	–0.0362 (2.00)**
Growth spline (0.75–2)	0.0099 (2.86)***	0.0051 (0.67)
Liquid assets	–0.0904 (30.39)***	0.1175 (13.84)***
Collateral	–0.1128 (33.52)***	–0.2868 (37.95)***
Inventories	0.0951 (25.16)***	0.1024 (10.72)***
Cost of sales	0.0103 (34.13)***	0.0093 (15.23)***
Size	–0.0246 (29.39)***	–0.0489 (24.68)***
Observations	151,584	83,113
Number of firms	33,190	21,839
R^2	0.07	0.06

Collateral, fixed assets/total assets; cost of sales: total cost of sales/total assets; inventories: finished and work in progress stocks/total assets; liquid assets, cash and liquid deposits/total assets; size, log of total assets.

(a) Table 5: performance splines.

Table 6
Age splines

Dependent variable	Trade credit/assets	Trade credit/total debt
Age spline (0–1)	0.0475 (4.85)***	0.0560 (2.96)***
Age spline (1–2)	–0.0019 (0.63)	0.0105 (1.66)*
Age spline (2–3)	–0.0033 (1.36)	0.0108 (2.12)**
Age spline (3–4)	–0.0036 (1.55)	0.0205 (4.23)***
Age spline (4–5)	–0.0042 (1.78)*	–0.0019 (0.39)
Age spline (5–6)	–0.0053 (2.25)**	0.0030 (0.62)
Age spline (6–7)	–0.0035 (1.47)	0.0055 (1.16)
Age spline (7–8)	–0.0035 (1.45)	0.0011 (0.24)
Liquid assets	–0.0985 (33.42)***	0.1171 (14.80)***
Collateral	–0.1346 (40.76)***	–0.3113 (44.43)***
Inventories	0.0824 (22.54)***	0.0885 (10.00)***
Cost of sales	0.0091 (31.48)***	0.0090 (15.24)***
Size	–0.0171 (22.97)***	–0.0308 (18.42)***
Observations	171,359	92,332
Number of firms	37,295	24,155
R^2	0.07	0.06

Collateral, fixed assets/total assets; cost of sales: total cost of sales/total assets; inventories, finished and work in progress stocks/total assets; liquid assets, cash and liquid deposits/total assets; size, log of total assets.
Absolute value of t-statistics in parentheses. All regressions include as control variables: firm-fixed effects and year dummies (not reported).
*, **, and *** significant at 10, 5, and 1%, respectively.

(b) Table 6: age splines.

8.5 Figure 3: Trade credit/assets versus age of the firm; Figure 4: Trade credit/total debt versus age of the firm

Read:

- Figure 3 shows that trade credit/assets rises sharply in the first year and then gradually declines.
- Figure 4 shows trade credit/total debt rising over early years and then remaining high.
- Together, the figures indicate that trade credit has a strong life-cycle pattern.

Economic message: Supplier-customer links take time to build. Trade credit becomes especially important once the relationship is established, but older firms eventually gain alternative finance and rely less on trade credit relative to assets.

Detailed interpretation: Figure 3 and Figure 4 are the best visual evidence for the relationship-based enforcement channel. They show that the role of trade credit changes as firms age.

(a) Figure 3: trade credit/assets versus firm age.

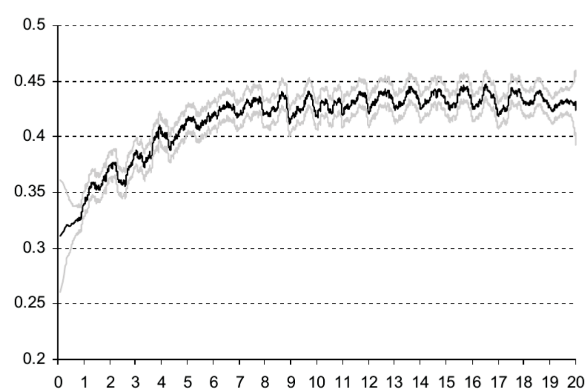


Figure 4

(b) Figure 4: trade credit/total debt versus firm age.

9 Short summary

- Trade credit is widespread and apparently expensive.

- Suppliers can enforce repayment better than banks because they can stop supplying inputs.
- Suppliers also provide liquidity insurance by tolerating late payments or expanding trade credit during shocks.
- High implicit interest rates include default and insurance premia.
- Firms with low collateral rely more on trade credit.
- Firms hit by liquidity shocks use more trade credit.
- Fast-growing and poorly performing firms both use more trade credit, producing nonlinear patterns.
- Firm age predicts trade credit, consistent with relationship building.
- The paper links real production relationships to financial contracting.

10 Conclusion

10.1 Core conclusion

Cunãt shows that trade credit can exist even with competitive banks because suppliers are not just lenders. They are input providers embedded in ongoing production relationships. This gives them enforcement power and creates incentives to insure customers against liquidity shocks.

10.2 Economic intuition

A supplier benefits from keeping a customer alive because the relationship produces future surplus. This makes the supplier willing to tolerate late payment or provide additional credit when a customer faces a temporary shock. A bank, which lacks the same production relationship, cannot enforce or insure in the same way.

10.3 Why trade credit is expensive

The high implicit interest rate on trade credit is not necessarily inefficient or exploitative. It reflects the supplier's funding cost, the probability of default, and the expected cost of liquidity support. When suppliers have higher funding costs than banks, the cost is amplified because trade credit is risky.

10.4 Empirical takeaway

The UK firm evidence fits the model. Trade credit is higher when collateral is low, when liquidity shocks arrive, when firms grow quickly or perform poorly, and when firm age suggests that supplier-customer relationships are forming.

10.5 Incremental contribution revisited

The paper's lasting contribution is to interpret trade credit as a relational financial contract. It shows that supplier finance is not an anomaly to be arbitraged away by banks; it is a contract that emerges because suppliers have information, enforcement, and relationship-continuation incentives that banks do not have.

10.6 Limitations and extensions

The evidence is not a randomized test of supplier relationships, and age is an imperfect proxy for relationship strength. Future work could use direct supplier-customer links, contract-level late-payment data, or shocks to supplier liquidity to test the theory more directly.